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Interoperability: Our exciting and terrifying Web3 future

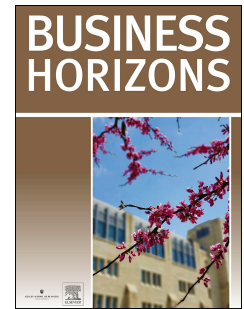
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**Interoperability:
Our exciting and terrifying Web3 future**

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Interoperability: Our exciting and terrifying Web3 future

Abstract

This article introduces the next major generational evolution of the web: Web3. We review the fundamental evolution of the internet and the web over the past three decades, including a brief presentation of the publications in Business Horizons that are important in a discussion of the emergence of Web3. We then discuss what these recent developments mean to organizations, consumers, and the public. Though the degree to which Web3 will be widely adopted is uncertain, these technologies are already creating both exhilarating and terrifying implications for e-commerce, digital media and online social networking, online marketplaces, search engines, supply chain management, finance, and more. We propose that technical, organizational, and regulatory interoperability will need to be considered and managed in order for Web3 to deliver on its promises of value, and that failure to consider these interoperability components may destroy economic value, consumer confidence, or social issues online. We also call on our fellow researchers to focus on these interoperability issues and how they might impact the bright and dark sides of Web3 technologies to help us understand and shape our Web3 future.

KEYWORDS: Web3; Blockchain; Metaverse; NFT; DeFi; Cryptocurrencies

"The Internet? I don't think we've even seen the tip of the iceberg. I think the potential of what the Internet is going to do to society both good and bad is unimaginable. I think we are actually on the cusp of something exhilarating and terrifying."

— David Bowie (1999)

1. Imagining the unimaginable

In a BBC interview during the height of the dotcom boom, David Bowie observed that through the internet, the actual context and the state of content was going to be very different to anything we could envisage. Although his interviewer could not, at first, comprehend that the internet was more than just a novel information delivery mechanism, Bowie was precisely right, and his predictions remain valid today: the internet and its applications continuously change in ways, both good and bad, that we cannot foresee or imagine. Approximately 25 years into the mass adoption of the internet, we find ourselves on the cusp of a major generational shift of the web. The impact of Web3, some argue, could be as big as the initial shift from analog to digital (Gartner, 2022), but what are some of the implications for organizations, consumers, and the public? In this article, our goals are to build on existing works that articulate what Web3 is (e.g., Murray et al., 2021) and to alert our readers to the exhilarating and terrifying future that lies ahead. We focus on the bright and dark sides of Web3 technologies, the organizational capabilities that will influence its use, and its overall impacts to help us understand and shape our Web3 future for good.

We begin by offering a brief overview of the three generations of the evolution of the web. Next, we use the argument of interoperability and the notion that managers ought to think about technologies and political environments differently if they are to succeed in the Web3 ecosystem. We conclude with a short discussion of the potential impacts, both good and bad, these developments may have on individuals, brands, organizations, and society as a whole.

2. The evolution of the web

2.1. "The Web" aka Web 1.0 – An information economy

The early days of the "information superhighway" were characterized by static, read-only webpages that were powered by early versions of HTML and enabled by data transmission protocols like TCP/IP, SMTP and HTTP. Collectively, these developments allowed users to transfer data between applications and devices on a network, send and receive emails, or read hypertext online (Choudhury, 2014). One major limitation of this first generation of the Web, which with hindsight we might now think of as Web 1.0, was that it was highly technical in nature. This served as a significant barrier to content creation on the internet (Cormode & Krishnamurthy, 2008). Another limitation was that there was no mechanism by which to monetize content on Web 1.0 (McCormick, 2021). As a result, the success of the early web was limited, and throughout the dotcom bust of the late 90s we learned that value-propositions of websites needed to be more clearly defined and developed.

2.2. Web 2.0 – A platform economy

Supported by HTML improvements and the development of XML technology, Web 2.0 overcame the web designer bottleneck, where only technically proficient people could make changes to static webpages. Now, anyone could consume *and* post (read and write) content to dynamic webpages, blogs, etc. (Kietzmann et al., 2011; Voshmgir, 2020). A platform economy developed, largely run by massive internet companies known today as the Big Five – Alphabet (Google), Amazon, Apple, Meta (Facebook) and Microsoft – alongside many smaller ones. These platforms acted as trusted intermediaries; they are so-called “walled-gardens” that tightly govern the enclosed system and control what users can and cannot do with their own data (e.g., photos, videos, posts) once on the platform. The data we generated, centrally stored on phones, computers, or decentralized in the cloud, was predominantly managed by third parties, and made accessible to us, anytime and anywhere for a cost (through membership fees or advertising exposure). Unfortunately, as with all technologies, the darker side of unintended consequences eventually also emerged (Baccarella et al., 2018; Demetis, 2020) – Web 2.0 had many risks (e.g., cybersecurity, privacy, bullying (Demetis & Kietzmann, 2021)) and produced high costs of document handling (e.g., disaster recovery, backups), as well as non-transparencies along the supply chain of goods and services (Montecchi, Plangger, & Etter, 2019).

But what if we start removing these walls and think about an internet without the notion of big platform providers altogether? Will this even be possible, and if so, will Bowie be correct again and “the interface between the user and the provider [will] be so *in sympatico* that it’s going to crush our ideas of what mediums are all about”?

2.3. Web3 – A token economy

Web 2.0 was a frontend revolution that allowed minimally tech-savvy users to create content online, and it opened up previously hidden communication channels. In contrast, Web3 (not to be confused with Web 3.0¹) is a backend revolution that focuses much less on changing the user interface and instead on replacing central data storage with data stored in a widely distributed manner. It is essentially a new way of organizing information. The next generation of the web, as opined by Ethereum co-founder Gavin Wood in 2014, will be built on the disintermediation of many of the existing parties involved in data governance. Instead of the current server-client relationship, in Web3, data will be stored on blockchains, in multiple copies on a P2P network.

[Insert Box 1 Here – This Placement Must Be EXACT]

On the consumer side, the value proposition of Web3 is easy to see. Several proposed versions of Web3 include rewarding users with a native token with each tweet, post, or video, instead of the current centralized monetization structure that contributes to the financial wealth of platform shareholders (CEOs, investors, employees). These tokens become the atomic unit of Web3. Although similar in nature to cryptocurrencies, these tokens can give token-holders specific ownership stakes, rights to vote, or ways to cash in. In other words, on Web3, everyone has the potential to monetize their own data. Tokens can represent anything from an “asset to an access right, such as gold, diamonds, a fraction of a Picasso painting or an entry ticket to a concert, or to reward social media contributions, incentivize the reduction of CO2 emissions, or even for watching an ad” (Voshmgir, 2020). By issuing and exchanging these tokens, the ensuing value

can be shared among more people, and people who do not trust each other online can come to binding agreements over the web – without the help of another party. These agreements can be executed automatically (Voshmgir, 2020) through smart contracts and decentralized applications (known as dApps). For instance, in real estate, the need for legal counsel or other advisory services between buyers and sellers could become less important, resulting in lowered costs.

Among incumbent organizations, the adoption of Web3 will take time, as the platform providers of today incrementally embed Web3 applications into their Web 2.0 presences and business models. Nike has already filed several new trademarks to make and sell virtual Nike-branded sneakers and apparel. Buyers of real-world apparel will be able to dress their avatars with the same clothes in games (e.g., Nike collaborates with Fortnite and Roblox already) and other apps. New entrants in spaces like distributed and decentralized finance (DeFi), virtual reality, and metaverses will disrupt the Web 2.0 status much more quickly. Decentraland, a Web3 version similar to SecondLife and other virtual social platforms, allows users to meet, play games, attend live events, trade in the marketplace, engage with brands, visit galleries etc. Their purchases, for instance for virtual plots of land, are registered as NFTs, paid via the MANA cryptocurrency and stored in a digital wallet. This Web 2.0 to Web3 trajectory is not unlike the transition from Web 1.0 to Web 2.0, where existing companies added social elements into their existing webpages, whereas new entrants built ‘social’ into every element of their business model and online presences from the start. Regardless, any scenarios related to the token economy (that is, no adoption, partial adoption, or full adoption of Web3) will introduce significant implications for e-commerce, digital media and online social networking, online marketplaces, search engines, supply chain management, finance, and more.

3. Web3 building blocks

To understand the evolution of the web and the impact that Web3 developments will have on current social, commercial, financial, political, and legal aspects (see Table 1), a quick summary of the fundamental building blocks is required.

[Insert Table 1 About Here]

Augmented Reality (AR) and *Virtual Reality (VR)* are the two Web3 topics that emerged first. *AR* superimposes digital data (e.g., sound, video, graphics, haptics, text) on top of our physical reality. Imagine a real estate app that displays data about a property (e.g., square footage, property tax) when a user looks at it through the lens of a mobile phone camera. In contrast, *VR* presents a complete 3D virtual representation of the actual world, or of a totally fictitious digital world (e.g. one can customise their own virtual supermarket with their favourite products and shop from their virtual corridors while wearing their VR headset). In our real estate example, a VR video of a property allows potential buyers anywhere to virtually walk around a house and get a realistic impression of the building and yard without ever visiting the site in person. Mixed reality combines aspects of the actual reality (the physical world around us) with the power of VR. For example, AI generated avatars that appear lifelike can interact with people in chatbots or to teach empathy to healthcare professionals. In *Business Horizons*, AR and VR articles that garnered significant attention focused on consumer or employee engagement (Scholz & Smith,

2016; Parise et al., 2016; Maltseva, 2020) and on business models and strategies (Hinterhuber, 2022; De Regt et al., 2020; Bermann & Polack, 2021).

Instead of relying on a central or decentralized approach to storing data (e.g., on organizational information systems or in the cloud), *Blockchain* is a technology that leverages a ‘distributed ledger’ to record transactions (Morkunas, Paschen, & Boon, 2019). The integrity of data on a blockchain is maintained by a peer-to-peer network of computers that verify each transaction before it is approved by consensus and subsequently added to the ledger. By safely recording transactions on the blockchain, incumbent institutions (e.g., banks, insurances) are no longer needed to broker online transactions. Although ‘the blockchain’ was initially developed to disintermediate banks (in conjunction with Bitcoin), the technology has since expanded to underpin a wider range of possible applications that run on blockchain (Hughes, Park, Kietzmann, & Archer-Brown, 2019). Smart contracts, for instance, rely on cryptographically signed transactions on the blockchain network. They automatically execute legally relevant actions (such as the transfer of tokens or payment of cryptocurrencies) according to the terms of a contract or an agreement (e.g., when an item is delivered, or a task has been completed). Blockchain has become a strong focus of research in *Business Horizons* since 2019, with key authors and articles contributing to what we know about supply chain resilience (Min, 2019), supply chain provenance (Montecchi et al., 2019), entrepreneurship and value creation (Chen, 2018; Malhotra, 2022; Sharma, 2021; Angelis & Da Silva, 2019), smart contracts (Larios-Hernández, 2017; Živković, 2021), cryptocurrencies (Chaudhry, 2017; Lee, 2019) and business in general (Kietzmann et al., 2019).

Non-fungible tokens were next in the timeline of technologies important for Web3. Commonly referred to as NFTs, these are digital certificates of ownership that secure digital files on a blockchain and turn original digital files into tradeable digital assets (Wilson et al., 2021). Whereas digital data can be infinitely replicated, the invention of NFTs can allow one to prove that they are the real ‘owner’ of a unique digital asset. Unlike other emerging and blockchain based digital assets such as cryptocurrencies, NFTs are so named because they are non-fungible: each NFT has a unique identifier which ensures that it is unique and cannot be replaced or traded with an NFT or copy of the same. NFTs are currently used as certificates of ownership related to digital art and collectables, decentralized finance, and in digital games and metaverses. Interestingly, they can be used to both dematerialize real world assets and to guide the materialization of virtual assets (e.g., through deeds for virtual properties or clothes for game avatars (Marder et al., 2019), signaling broad and versatile use (Wilson et al., 2021). NFTs are a recent but very important development, with few *Business Horizons* publications so far. Wilson et al. present an ecosystem view of NFTs (2021), and Chohan and Paschen share a marketing perspective related to NFTs (2021).

The most recent addition, although by no means the least impactful one, is the notion of a *metaverse*, where all the above developments come together. It refers to an immersive online world where users can interact with each other and their environments through virtual and augmented reality technologies. Following the October 2021 announcement that Facebook is rebranding itself as Meta Platforms, Inc. – a move driven by the organizations envisioned future of the metaverse – the term metaverse has been garnering increased attention. Zuckerberg describes the metaverse as the next evolution in social technology, in which the user experience

of social media will move beyond 2D screens to become immersive, and in which users will interact in 3D spaces using holograms or avatars of themselves (Zuckerberg, 2021), and engage in commercial exchanges using NFTs and cryptocurrencies.

[Insert Box 2 Here – This Placement Must Be EXACT]

4. Interoperability requirements for Web3

As these technologies individually advance and collectively form Web3 applications, discussions surrounding how to prepare for such a future become more and more important. What capabilities will be needed to leverage Web3 opportunities for improved social and economic outcomes and mitigate Web3 risks? Specifically, we argue that *interoperability* in the Web3 ecosystem will play a critical role in the emergence of new firms, applications, and modes of thinking.

Interoperability matters for a number of reasons. At a high level, the term suggests that systems, devices, and organizations and their “solutions” can connect with each other and engage in meaningful data exchange without demanding additional user involvement. In a Web 2.0 environment, this appears comparatively straightforward, where we can communicate with each other through messages and emails regardless of operating system, devices, or service providers, and where we use a common currency and payment mechanisms (e.g., credit cards) when we purchase goods and services online. Had these and other technical and institutional choices been made differently, the mass adoption of Web 2.0 would have been hard to imagine. Web3 still suffers from these and other challenges, though. For instance, scaling Web3 to maturity requires blockchains to talk to each other (they currently do not) so that they can support collaboration and the transfer of digital assets (e.g., transferring and trading NFTs across networks), of monetary value (using a single wallet or smart contracts between applications) and of data (e.g., cross-game, interoperable avatars, or health and patient profiles that can be used everywhere), among other areas.

When we consider the importance of interoperability and its constituent elements of technological, organizational and regulatory interoperability (see Figure 1), we will see new Web3 applications, products and services, particularly those that connect previously disconnected industries and sectors, emerge and flourish.

[Insert Figure 1 About Here]

4.1. Technological interoperability

In a technological sense, Web3 and its constituent applications are heavily reliant on participants and the network effects they generate, but are currently impeded by nonintegrated technologies, especially across many different blockchain environments. For example, a holder of Bitcoin cannot yet easily use their Bitcoin to transact with a party who primarily resides in the Ethereum community, much less use their Bitcoin to activate an Ethereum smart contract. In fact, blockchain interoperability has been a persistent issue since the emergence of various blockchains after Bitcoin (Belchior et al., 2021). Blockchain participants recognize that the existing fragmentation of users across the popular blockchains are encumbering the overall

growth of the ecosystem. Starting with the introduction of Ethereum in 2015 as a viable competitor to Bitcoin, dozens of other blockchains now have appreciable usage with each making specific decisions and tradeoffs related to decentralization, security, and scalability. For example, while Bitcoin is the most valuable, public blockchain today, several private companies such as UnitedHealthcare and Walmart have experimented with their own private blockchains. Furthermore, Ethereum is in the process of migrating to a fundamentally different consensus architecture from Bitcoin, ostensibly taking a risk on security and decentralization to improve scalability. While each of these blockchains offer different advantages to consumers seeking specific feature sets, the fragmented nature of blockchain usage, with various ‘camps’ purporting the superiority of their chains, acts as an obstacle to the growth of the Web3 ecosystem.

Many attempts have been made to build middleware products that allow inter-blockchain communication and benefit from increased network effects. Polkadot, Loom Network, Cosmos, Plasma, and Lisk have tried to address the current state of limited interoperability across blockchains. All these solutions make tradeoffs regarding security and scalability to enable interoperability. For example, each of the aforementioned interoperability projects enable cross-blockchain communication by acting as a translation layer. However, to use this translation layer, users must acquire cryptocurrency unique to that project. Critics suggest that these projects are simply offering interoperability by siphoning users to their own platforms and undermining the blockchain decentralization philosophy. The technical interoperability problem has yet to be solved, and it will be interesting to see which standard ultimately facilitates the emergence of Web3.

4.2. Organizational interoperability

Creative ways to reconfigure value chains to minimize zero-sum and erosive industry architectures will need to be explored. In this manner, organizational interoperability is a critical element that needs to be considered alongside technological interoperability. Organizations must be willing to adopt the decentralized philosophy of Web3, and a vibrant network of firms and service providers must collaborate or coopt to pave the way for integrated products and services that will in turn enable the growth of the Web3 ecosystem. This includes organizations such as software development shops, legal firms, marketing firms, hardware manufacturers, network specialists and many more.

Radical applications of Web3 and blockchain, such as a reconfiguration of our healthcare system where data is stored not in the private databases of health records companies, but on a public ledger (Reegu et al., 2022), require substantial buy-in from organizations along the industry value chain. This includes incumbents like Epic and Cerner, hospitals, insurance companies, diagnostic laboratories, clinics, pharmacies, and numerous other stakeholders. Another transformative Web3 idea that has been proposed is peer-to-peer mortgage loans, a subset of a broader set of applications known as Decentralized Finance, or DeFi (Sriman & Kumar, 2022). It has been suggested that with Web3, traditional intermediaries like banks and mortgage brokers will be obviated and that disinterested and untrusting parties can provide loans to each other, and debt will be serviced and enforced through automated smart contracts. It is unlikely traditional banks, mortgage lenders, real estate agents, lawyers, and other existing stakeholders will simply move aside to allow this phenomenon to supplant them; these stakeholders will need to participate and, in many cases, mutually benefit from the growth of Web3.

4.3. Regulatory interoperability

Once technological and organizational interoperability issues are addressed, regulatory and political entities that govern our social and economic exchanges also need to be involved. Web3 proponents must work with regulators to ensure that there is enough regulatory oversight and support to ensure a broad base of consumers are not defrauded, which in turn will attract more Web3 participants and enhance network effects. But the oversight cannot be so draconian that innovation is discouraged or punished.

Extending our examples above, our traditional healthcare or financial infrastructures will need to be on board with such a radically disruptive paradigm, and individuals and organizations who wish to participate in this new paradigm will need to account for the resource and time investments required to lobby and work with these key stakeholders. Some industries are more highly regulated than others. For example, decentralized finance applications in the US will undoubtedly require governance by the Securities and Exchange Commission. Conversely, decentralized gaming with few tradeable assets will likely not need as much regulatory approval. Overall, regulatory interoperability (i.e., the inclusion and blessing of jurisdictional authorities) is an important consideration in determining the viability of any Web3 application. While it may first appear that regulators may suppress innovation in this space through overly oppressive watchdog-like practices (as seen in the Chinese's government's decision to outlaw the use of cryptocurrencies completely), the tide is turning in the general blockchain community and even Bitcoin maximalists like Michael Saylor (former CEO of Microstrategy) argue that a healthy amount of regulatory oversight will improve consumer confidence and actually attract more users to the blockchain community.

Each of the three interoperabilities is essential but by itself not sufficient for the success of Web3 applications. If all three interoperability pieces are considered and optimally managed, profoundly transformative Web3 applications can generate significant economic, social and humanitarian value beyond Web 2.0 applications of today. However, if these interoperability components are subverted, ignored, or mismanaged, Web3 outcomes may destroy economic value, consumer confidence and further worsen existing Web 2.0 social issues.

5. Our interoperable future: Exhilarating and terrifying

As organizations grapple with these new considerations of interoperability, Web3 is becoming more and more of a reality for early movers. Brands like Nike and Gucci are investing in metaverses, and Crate & Barrel recently announced a senior executive role for the metaverse (i.e., Sebastian Brauer is the new Senior Vice President and Head of Metaverse, Web3). Nadella, Microsoft's CEO, claims that the company is leading innovation in the "industrial metaverse" platform through their work on IoT, digital twins, mixed reality applications. While some are highly critical of Web3 developments and speculate that "it's all consumerism and little impact" (Glassenberg, 2022), others celebrate the effect it already has on even the most unlikely industries. For instance, through low-risk medical training, healthcare practitioners can practice procedures together, collaboratively, on a virtual patient over Zoom. Medical AR uses can allow surgeons to see CT and MRI data 'inside' of the patient as they perform a procedure, too.

With experts on both sides of the argument, lots of questions emerge about the actual value of Web3. The answers depend on the degree to which interoperability is observed in Web3 developments. In Bowie's terms, "we are actually on the cusp of something exhilarating and terrifying" (see Figure 2).

[Insert Figure 2 About Here]

5.1. Exhilarating: Web3 opportunities

If we have learned anything from the first iterations of the web, it is that its evolution, and the larger digital ecosystem in which it exists, are undeniably exciting. The positive possibilities of Web3 are wide ranging, including the following four opportunities:

5.1.1. Redistribution of power

One exhilarating possibility of Web3 is the potential disruption of the status quo that could occur by eroding the power of the Big Five Web 2.0 gatekeepers – Meta (Facebook), Amazon, Alphabet (Google), Apple, and Microsoft. Each of these tech giants functions as a virtual monopoly, with the overall results being centralized power and limited competition. As the reliance on Web 2.0 legacy platforms decreases, smaller start-up businesses will fill the void, offering consumers more choices and more freedom. The opportunities arising will be wide ranging, including opportunities for research firms on Web3 consumer behaviour, experiences, and wants, to new products or services, new intermediary functions, among others.

5.1.2. Decentralization and fairer governance

Instead of a few platforms creating private networks for public infrastructure (e.g., AirBnB, Twitter, Uber, etc.), Web3 promises open networks that share value across multiple providers. This holds a number of promises. Today's internet infrastructure is susceptible to broad outages and hacks. The decentralized nature of blockchains removes single point failures and is expected to be more reliable. In addition, Big Tech's outsized influence on internet regulations (e.g., net neutrality) may also be confronted. To date, governments have had limited success regulating firms and a shift towards decentralization may open opportunities for regulatory reform.

5.1.3. Increased consumer choice

Consumers currently have no choice but to engage with these platforms if they want to participate in Web 2.0. Importantly, they often do so without realizing the amount of data they are implicitly sharing or the ways in which their data is used. Web3 offers consumers the ability to subvert the Big Five and regain ownership of their data – something that would likely be appealing to many, as consumers today already show a high appetite for subverting organizations (Wilson, Robson, & Pitt, 2021). A shift to Web3 could be yet another shift in the balance of power between organizations and consumers (see: Wilson, 2018; Wilson, Robson, & Botha, 2017), with consumers being the beneficiary. Consumers will decide what information they share, and with whom. And because this data is still valuable to marketers, consumers, not companies, will be compensated, likely using smart contracts that 'uncomplicate' and automate this process. Ownership of data has implications for creative works as well and will afford artists more autonomy over how their intellectual property is shared and monetized.

5.1.4. Equal access

Web3 is already impacting users in developing parts of the world, where marketers have long needed to engage in creative and unique tactics to serve customers (Beninger & Robson, 2015). Access to capital is difficult for low-income individuals due to traditional banking barriers, but as more financial products are offered on blockchain technology, new opportunities arise to increase financial inclusion. DeFi products, including lending/borrowing, payments, insurance, savings and investing, are giving millions of base-of-the pyramid consumers the means to open accounts, start businesses, and participate in the global economy. Services that once required a financial institution, such as global remittance payments, are being completed faster and at a fraction of the cost. Furthermore, cryptocurrencies are shielding citizens from corrupt and damaging actions of their governments and central banks, which can lead to inflation, devaluation of currencies, and restricted access to markets.

5.1.5. Summary

Web3 applications hold the promise not just to enhance creativity and imagination, but as the healthcare example shows, can also help people overcome real world obstacles through training, upskilling, education, exploring new career opportunities etc. in metaverses. For the bright side of Web3 to be realized, technological, organizational and regulatory interoperability must be enabled and appropriately managed. Just as exhilarating as these developments promise to be in practical terms are, so are they in terms of research opportunities. We encourage researchers from all disciplines to work on these and many other potentially very progressive Web3 topics.

5.2. Terrifying: Web3 risks

If any of the three components of interoperability are subverted, we may see a much darker side of Web3. For example, as soon as Meta announced their commitment to the Metaverse, voices became increasingly frequent and loud about how such a move could indeed be terrifying, including the following four risks:

5.2.1. Concentration of power

The first general bucket of criticism levied at Web3 is that it does not – and possibly will never – deliver on its philosophical promises. The issues of centralization, transparency, and power imbalances, the correction of which is at the heart of Web3 design, must be mitigated through interoperability. Currently, however, there are a limited number of exchanges to buy and sell coins, and even fewer for NFTs, resulting in a natural trend towards centralization and concentration of power. Even still, nearly all NFT activity is happening through a few vendors in the Ethereum ecosystem. If the early days of Web3 have revealed anything, it is that inherently decentralized systems do not inevitably lead to decentralized markets or more evenly distributed power. The increasing talk around heavily centralized, government-backed cryptocurrencies such as the Federal Reserve's digital dollar (i.e., FedCoin) concerns many as being antithetical to the blockchain ethos. In this case, regulatory interoperability without commensurate technological interoperability portends the emergence of a terrifying Web3 application. Transparency, another fundamental characteristic of Web3, is also seen as a hollow promise by some, given the lack of recourse available to individuals in the event of a security breach. What purpose do publicly available ledgers serve if anonymous hackers can steal coins (irreversibly) for all to see?

5.2.2. Decentralization woes

In another example of the need for technological interoperability, compared to Web 2.0 systems, Web3 platforms are notoriously inefficient and slow. For example, Bitcoin's transaction processing capacity is approximately 3-7 transactions per second (TPS), while Ethereum's capacity is 15-25 TPS (Craig, 2021). Comparatively, VISA regularly processes 1,700 TPS (VISA, n.d.). Slow processing times are manageable at current usage rates; however, they present a clear scalability challenge for blockchains. This issue is compounded by the fact that proposed improvements to processing rates, (mainly foregoing 'proof of work' consensus mechanisms), erode decentralization, a core tenet of Web3. Performance and scalability are issues that must be addressed to convince the public that the era of Web 2.0 is nearing an end. The environmental impacts of blockchains are also significant in terms of both hardware and power requirements. Over the past decade, there has been an arms race to acquire larger processing rigs, HVAC equipment, hardware engineers, and support systems necessary to execute work on blockchains. And because of the winner take all nature of this work, a massive amount of redundant work is performed and discarded by these rigs every day. In addition to the hardware requirements, the computing power needed for 'proof of work' consensus algorithms used in many mining and processing cryptocurrency transactions requires significant energy. A recent report calculated that a single Bitcoin transaction uses 1,173 kWh, at an average cost of USD \$173 (Crypto Energy Consumption, 2021). Given the significant energy requirements of Web3 technologies, some have advocated for 'proof of stake' consensus mechanisms, which use less energy per transaction (Tante, 2021). In addition, many blockchains, including those for Bitcoin and Ethereum, have pledged to use more renewable sources of electricity. Despite this progress, the environmental footprint of Web3 continues to be worrisome.

5.2.3. Consumer exploitation

Another critique of Web3 technologies is that they are simply a grift. This criticism stems from the limited supply design of cryptocurrencies, which creates a zero-sum system (Hanley, n.d.): for every winner there is a loser because the number of coins is fixed. Today, many people treat crypto as an investment opportunity, a speculative asset, since most coins are too volatile to be used like fiat money. The problem arises when an investor tries to extract actual money and needs to find someone willing to buy the coins for more than they purchased them. Critics argue that products stemming from the Web3 environment, including NFTs, Play to Earn Games, Metaverses, and Decentralized Autonomous Organizations (DAOs) exist solely to get people to buy cryptocurrencies, and allow early adopters to cash out (Tante, 2021). As a result, Web3 evangelists are seen as unreliable sources, as they have financial stakes in an intangible development whose value is based on public perception. Crypto communities tend to be insular groups that are resistant to skepticism, as any negativity can have a chilling effect on their investment. Doubt is aggressively policed in these communities resulting in a feedback loop of positivity and an 'in-group' dynamic, complete with rituals and slang (e.g., "To the moon", WAGMI [we're all going to make it], HFSP [have fun staying poor], etc.) By default, any skeptic of Web3 is just a sore loser, who wishes they got in earlier. This dynamic is why some characterize NFT markets as casinos, detached from economic theory (Martin, 2021). This is further evidence that some regulatory interoperability will likely be a benefit to Web3, and act as an important deterrent for such exploitative applications.

5.2.4. A deeper digital divide

Many warn that Web3 will harm our ability to separate the ‘virtual’ from the ‘real’ and thereby distort how users perceive things in their real lives. Highly regarded VR, AR, and AI pioneer Louis Rosenberg, for instance, argues that Web3 will “exacerbate the existing problems of the Internet to a dangerously high level” and that Web3 could be, in social terms, far worse and divisive than Web 2.0 social media platforms (2021). Although promises for base of the pyramid countries are real, so are the risks that those who cannot afford the necessary hardware will be excluded, which is particularly worrisome when critical information appears more in metaverses than in more accessible media (radio, TV, newspapers, and Web 2.0 sites) (Sarwar, 2021). A deeper digital divide might emerge, highlighting the need for strong interoperability, consisting of a vibrant ecosystem of numerous vendors, suppliers, purchasers, and other participants along the industry value chain.

5.2.5 Summary

As we envision our collective Web3 future, we need to be aware of its negative potential and the startling developments that can emerge if we do not pay enough attention to technological, organizational and regulatory interoperability. Much like the exhilarating opportunities, these risks offer many research opportunities, and we encourage researchers from all disciplines to work on these and many other potentially very distressing Web3 topics.

6. A call to action

In the early days of Social Media, researchers (including the authors of this article) focused exclusively on the bright side. In hindsight, we know that this was not only overly optimistic, but also largely misleading. Had many negative developments (e.g., online defamation, bullying), been researched and discussed sooner, they could have been addressed earlier and led to better outcomes. It is against this backdrop that we encourage researchers to explore and discuss interoperability topics as well as the positive and potentially destructive social and organizational consequences of Web3. We organize a number of potential research areas and questions around these themes in Table 2 below, linking each question to recent works which could serve as informative foundations for future works.

[Insert Table 2 About Here]

7. Conclusion

The possibilities and implications of the emerging next generation – Web3 – are simultaneously exhilarating and terrifying. They range from the decentralization of power currently held by big tech companies to new user experiences and the potential for users to monetize their own data to issues related to inefficient processing and environmental damage hanging in the balance.

At this early stage, it is impossible to predict what the future of Web3 holds or which exhilarating or terrifying possibilities will come to fruition. We hope that researchers will pay attention to the many difficult interoperability challenges (technological, organizational, and regulatory), and move our understanding of Web3 developments, both good and bad, forward in

important ways.² As Web3 matures from a concept to a reality, we suggest that managers pay close attention to these positive and negative potentials so that we can collectively and consciously shape our Web3 future – however unimaginable it might be at this point.

² We would like to conclude with a shoutout to Alex Murray, Dennie Kim, Jordan Combs who, as we finished this call-to-action, published an article entitled “The promise of a decentralized Internet: What is web 3.0 and HOW can firms prepare?”. Congratulations on a wonderful and very timely article.

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Box 1: Web3 vs. Web 3.0

The notion of **Web 3.0** is very different from our conversation about **Web3**. First conceptualized by Tim Berners-Lee in 1999, Web 3.0 was going to be developed around creating a ‘Semantic Web’, with common data formats and exchange protocols that would make all internet data machine-readable. This would provide contextual and conceptual understanding and meaning to users across different content and information applications and systems. As Berners-Lee stated in 1999: “I have a dream for the Web [in which computers] become capable of analyzing all the data on the Web – the content, links, and transactions between people and computers. [...] Day-to-day mechanisms of trade, bureaucracy and our daily lives will be handled by machines talking to machines.” This notion remains largely unrealized, and only a small percentage of websites ever contained semantic web markup.

Box 2: Open vs. Closed Metaverses

A key distinction in the conceptualization of a metaverse is between ‘open’ and ‘closed’ forms. An open metaverse consists of a collection of interconnected worlds in which users have permissionless access to contribute to the environment. Ownership of digital items is independent of any platform allowing users to communicate and transact seamlessly across any world. Conversely, a closed metaverse operates within a company’s ‘walled garden’ and is only accessible to users who have been granted access. Content that is created in a closed metaverse remains within that world. Advocates for the open metaverse warn against a single entity (e.g., Meta Platforms) wielding too much power, however closed metaverses offer more privacy and security and may be more suitable for training and simulation applications.

Table 1. The evolution of the web

	The Web (Aka Web 1.0)	Web 2.0	Web3
Origin	Tim Berners-Lee 1989	Tim O'Reilly 2004	Gavin Wood 2014
Web Presences	Organization focused through homepages	Community focused through social networks, blogs, wikis	Individual focused through AR, VR, metaverses
Architecture	Dedicated infrastructure	Cloud computing	Edge computing and blockchain
Possible Content Control	Internet Service Provider (ISP) dependence	Platform dependence	Provider and platform independence
Information Flow	Static, read only, one-way	Dynamic, many to many	Experiential, livestreams
Society	Digitally represented through online accounts	Digitally interconnected through profiles on social networks	Digitally re-created through interoperable avatars
Politics	e-Government (1-way information Government- to-Citizen)	e-Participation, interactive e-services for citizens	Distributed e- governance, u-Government (Personalisation)
Law	e-contractual agreements	Intellectual property on the web, data protection, copyright in 3 rd party material (e.g., Creative Commons)	Smart contracts, Blockchain-based arbitration systems, algorithmic compliance
Economy	e-Commerce (1-way) agreements, Major national banks	Marketplaces, P2P transactions, sharing economy, digital banks	NFTs, Cryptocurrencies, Decentralised Finance
Advertising	Banner advertising and page views	Interactive Advertising and pay per click	Behavioral Advertising and engagement

Table 2. Research areas

Research theme	Research areas or questions
Regulatory Interoperability	• What are the optimal approaches to platform governance and how can these be implemented? (See: Chen, Pereira, & Patel, 2021; Lumineau, Want, & Schilke, 2021; Murray et al., 2021; Schmeiss, Hoelzle, & Tech, 2019). • How might regulations maximize social benefits such as inclusion (See: Chen & Bellavitis, 2020)?
Organizational Interoperability	• How can organizations utilize internal metaverses for training and simulation applications? • What training, organizational cultures, or employee development programs are best suited to fostering organizational interoperability?
Technological Interoperability	• How can consensus algorithms on blockchains be designed to limit energy use and resources? • How do consumers evaluate technological trade-offs in blockchains and/or middleware (e.g., transparency, security, scalability, etc.)? What key value drivers motivate consumer choice of blockchains and/or middleware? • How does Web3 transform human-technology interactions (See: Murray, Rhymmer, & Simon, 2021)?

Figure 1. Interoperability requirements for Web3

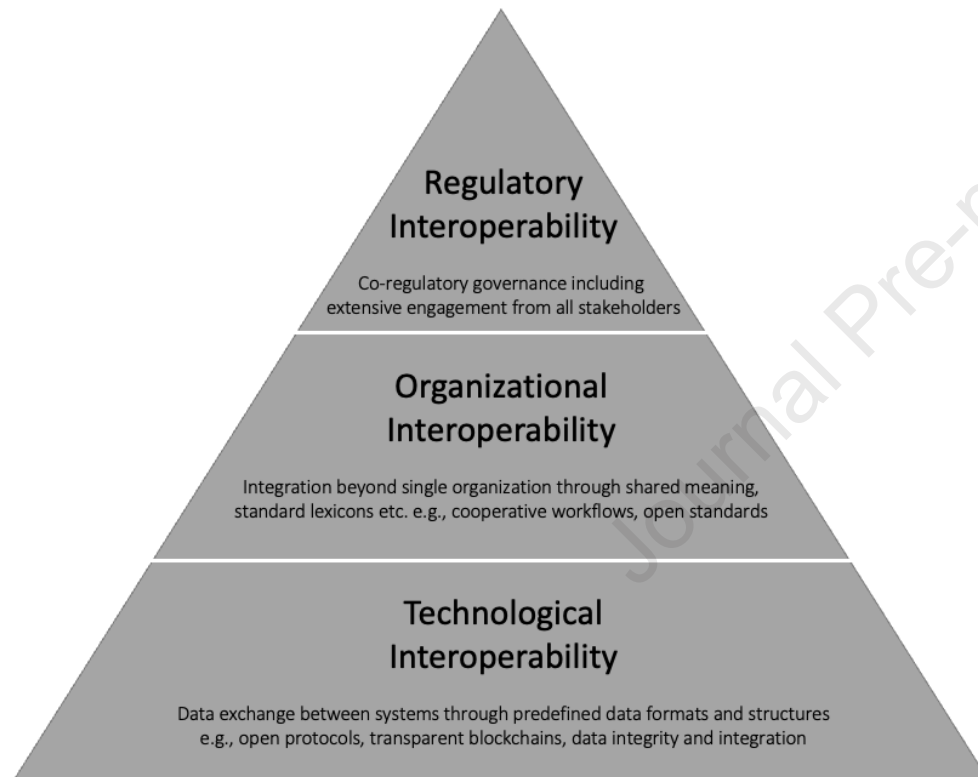
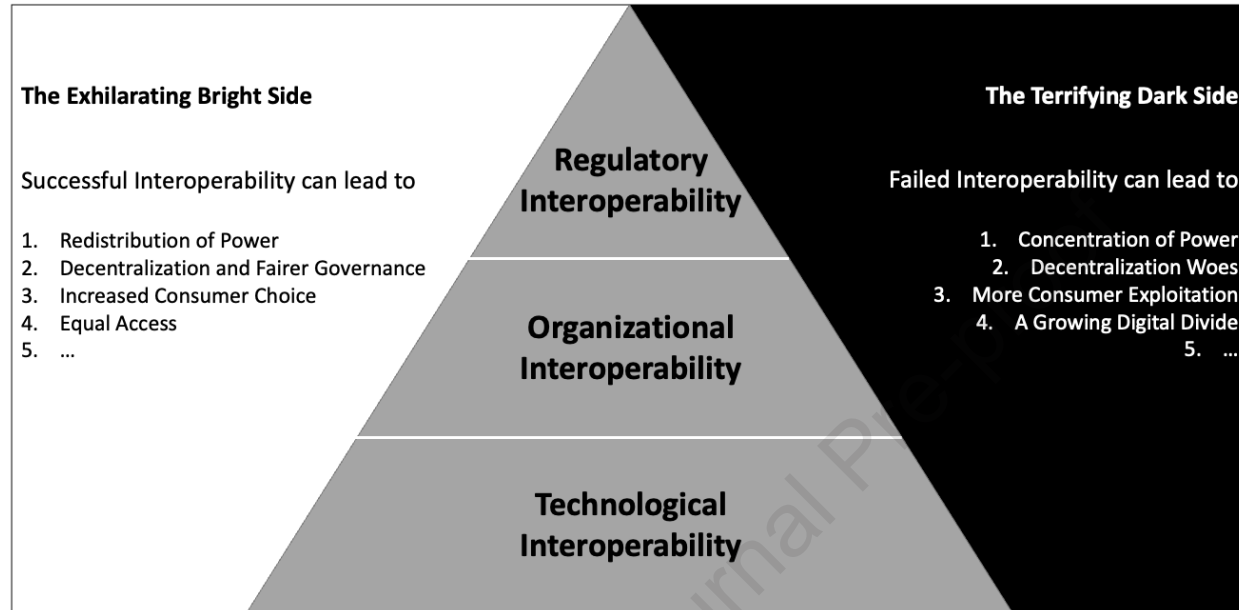


Figure 2. Interoperability, exhilarating, and terrifying Web3 outcomes



	The Web (Aka Web 1.0)	Web 2.0	Web3
Origin	Tim Berners-Lee 1989	Tim O'Reilly 2004	Gavin Wood 2014
Web Presences	Organization focused through homepages	Community focused through social networks, blogs, wikis	Individual focused through AR, VR, metaverses
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Possible Content Control	Internet Service Provider (ISP) dependence	Platform dependence	Provider and platform independence
Information Flow	Static, read only, one-way	Dynamic, many to many	Experiential, Personal lifestreams
Society	Digitally represented through online accounts	Digitally interconnected through profiles on social networks	Digitally re-created through interoperable avatars
Politics	e-Government (1-way information Government- to-Citizen)	e-Participation, interactive e-services for citizens	Distributed e-governance, personalized u-government
Law	e-contractual agreements	Intellectual property on the web, data protection, copyright in 3 rd party material (e.g., creative commons)	Smart contracts, blockchain-based arbitration systems, algorithmic compliance
Economy	e-Commerce (1-way) agreements, major national banks	Marketplaces, P2P transactions, sharing economy, digital banks	NFTs, cryptocurrencies, decentralised finance
Advertising	Banner advertising and page views	Interactive Advertising and pay per click	Behavioral advertising and engagement